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import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf

from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout # Layers
from tensorflow.keras.utils import to_categorical

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Preprocessing
X_train = X_train.reshape(-1, 28, 28, 1)
X_test = X_test.reshape(-1, 28, 28, 1)

X_train = X_train / 255.0
X_test = X_test / 255.0

y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Build CNN model
model = Sequential()

model.add(Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)))
model.add(MaxPooling2D(2,2))

model.add(Conv2D(64, (3,3), activation='relu'))
model.add(MaxPooling2D(2,2))

model.add(Flatten())
model.add(Dense(128, activation='relu'))
model.add(Dropout(0.5))

model.add(Dense(10, activation='softmax'))

# Compile model
model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

# Train model
history = model.fit(X_train, y_train,
                    epochs=10,
                    batch_size=64,
                    validation_split=0.1)

# Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)

print("Test Accuracy:", test_acc)

# Plot accuracy
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.legend(['Train', 'Validation'])
plt.title("Accuracy")
plt.show()

# Prediction
import random

index = random.randint(0, len(X_test)-1)
sample = X_test[index]

plt.imshow(sample.reshape(28,28), cmap='gray')
plt.show()

prediction = model.predict(sample.reshape(1,28,28,1))

```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 — 0s 0us/step

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Epoch 1/10

844/844 — 55s 63ms/step - accuracy: 0.9207 - loss: 0.2559 - val_accuracy: 0.9830 - val_loss: 0.0601

Epoch 2/10

844/844 — 48s 57ms/step - accuracy: 0.9739 - loss: 0.0882 - val_accuracy: 0.9880 - val_loss: 0.0409

Epoch 3/10

844/844 — 79s 54ms/step - accuracy: 0.9811 - loss: 0.0651 - val_accuracy: 0.9883 - val_loss: 0.0382

Epoch 4/10

844/844 — 81s 52ms/step - accuracy: 0.9842 - loss: 0.0530 - val_accuracy: 0.9893 - val_loss: 0.0376

Epoch 5/10

844/844 — 83s 54ms/step - accuracy: 0.9873 - loss: 0.0432 - val_accuracy: 0.9912 - val_loss: 0.0306

Epoch 6/10

844/844 — 44s 53ms/step - accuracy: 0.9890 - loss: 0.0375 - val_accuracy: 0.9908 - val_loss: 0.0349

Epoch 7/10

844/844 — 81s 52ms/step - accuracy: 0.9895 - loss: 0.0337 - val_accuracy: 0.9905 - val_loss: 0.0366

Epoch 8/10

844/844 — 46s 55ms/step - accuracy: 0.9905 - loss: 0.0293 - val_accuracy: 0.9915 - val_loss: 0.0341

Epoch 9/10

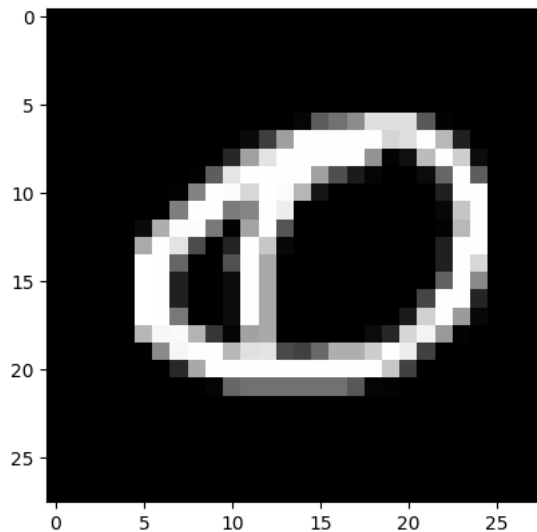
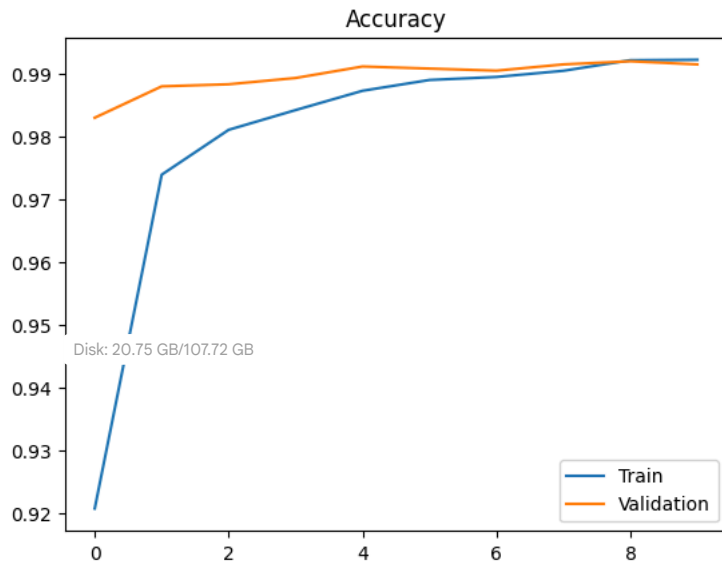
844/844 — 44s 52ms/step - accuracy: 0.9922 - loss: 0.0247 - val_accuracy: 0.9920 - val_loss: 0.0316

Epoch 10/10

844/844 — 44s 52ms/step - accuracy: 0.9922 - loss: 0.0236 - val_accuracy: 0.9915 - val_loss: 0.0356

313/313 — 4s 13ms/step - accuracy: 0.9909 - loss: 0.0282

Test Accuracy: 0.9908999800682068



1/1 — 0s 90ms/step

Predicted: 0

Disk: 20.75 GB/107.72 GB